Expecting the Expected? Documenting Nostalgia Effects on Baseball Card Prices

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Abstract

Identifying anomalies, or empirical facts that contradict theoretical predictions, is a common form of proof by contradiction in economics and finance. We identify an apparent pricing anomaly in the market for trading cards: large and persistent death effects of about 20% are observed following the death of legendary baseball players. Rational investors should anticipate each player's eventual death, so these results suggest that profit opportunities are being left on the table. The arguably causal estimates come from interrupted time series models that adjust flexibly for time pre- and post-death and the time of sale using high-frequency sales data. The effect is largest in the days immediately following the death, but prices remain elevated for at least 60 days. Generally, our findings reaffirm the presence of a causal "nostalgia effect" of celebrity deaths on consumer demand for collectibles and products associated with the celebrity.

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1. INTRODUCTION

Numerous studies in economics and finance identify anomalies, or empirical facts that contradict theoretical predictions (e.g., Schwert 2003; Thaler 2012). Identifying anomalies can provide important guidance to improve our theories of how people make decisions and, on the practical side, identify profit opportunities. The efficient markets hypothesis (EMH), a special case of rational expectations, is an important and relatively well-known theory that has received much empirical scrutiny (Fama 1970, 1998; Malkiel 2003).

Intuitively, the EMH suggests that asset prices should follow a random walk, as the asset's price at any moment in time reflects all publicly available information about the asset. Unsurprisingly, the straightforward predictions of the EMH have been tested in a variety of settings, typically yielding mixed results. On the one hand, several specific pricing anomalies such as day-of-week and month effects have been identified, suggesting that certain price movements are predictable (French 1980; Thaler 1987; Schwert 2003). On the other hand, the importance and permanence of these pricing anomalies are unclear, as they do not always replicate across time or markets and do not appear to be exploited by actively managed funds (Malkiel 2003; Schwert 2003).

We provide a loose test of the EMH and rational expectations in the unique, quickly growing market for trading cards, which represent an alternative investment class. Fueled by the pandemic, trading card sales on the online auction house Ebay increased by 142% in 2020.[2] The value of individual cards is on the rise as well: since 2016 the CL50, an index of 50 prominent cards across all sports (and a few card games, like Pokémon) thought to represent the trading card market, is up more than 360% as of January 2024, even after a large correction in 2022 and

[2] See https://www.ebayinc.com/stories/news/ebays-2021-state-of-trading-cards-report-spotlights-collecting-trends-and-industry-predictions/.

2023 from the 2021 peak.[3] The current study's focus is on the vintage (postwar) baseball card market. Card Ladder's indexes tracking the baseball and post-war vintage card markets are up 321% and 329%, respectively, since January of 2016.[4] Given the large run-up in prices of these collectible pieces of cardboard and the increasing sophistication of the market, an open question is how information shocks and collectors' expectations affect trading card prices – and how rational collectors' responses are.

Specifically, we test for the presence of a "death effect" on sports card prices following a player's death, which has previously been dubbed a nostalgia effect (Matheson & Baade 2004). The EMH suggests that because death is, at the time of writing, inevitable, each player's inevitable death should already be reflected in market prices. However, at least anecdotally, the idea that players' deaths affect trading card prices is well known amongst investors and collectors (e.g., Vintage 2020). We provide arguably causal estimates of a large and persistent death effect on the market prices of several legendary baseball players' rookie cards using an objectively defined sample and an interrupted time series (event study) research design that adjusts for broader trends in the trading card market.

On average, graded Topps rookie cards of nine Major League Baseball Hall-of-Famers increased by an average of 20% following their death.[5] This death effect is statistically significant and robust to a variety of modeling decisions and sample restrictions. Importantly, while the price increase is greatest in the days immediately following a player's death, significantly higher prices remain for at least 60 days post-death. These findings refute the idea

[3] The CL50 is tracked by Card Ladder: https://app.cardladder.com/indexes/cl50index. Accessed January 14, 2024.
[4] https://app.cardladder.com/indexes. Accessed January 14, 2024.
[5] Being elected to the Hall of Fame is the greatest honor a former baseball player can receive. Less than 1% of players receive this honor. See https://baseballhall.org/ for additional information.

that the trading card market is efficient and instead suggest that collectors and investors routinely react to information shocks that should have been fully anticipated.

In this sense, the current study follows a broad practice in economics that studies collectibles markets because they provide an accessible and ideal setting in which to test theoretical predictions by observing real, high-stakes decisions in the field (e.g., List 2002 a, b). It also contributes to a broad social science literature that investigates the sociological, psychological, and financial reasons that humans collect (or invest in) things that have little to no intrinsic value, which is an interesting question in its own right (e.g., Burton & Jacobsen 1999; Gérard-Varet 1995; Spaid 2018). Our findings also connect to the theory of rational inattention, which assumes that investors choose to consider a subset of the information available to them, and to ignore the rest (Sims 2003; Mackowiak et al. 2023), as the news coverage following a player's death may increase the salience of information regarding their greatness and collectability relative to that of other players. We briefly review the relevant literature in section 2. Sections 3 and 4 describe the data and econometric model, respectively. Section 5 presents the empirical results and Section 6 concludes.

2. Background and Literature Review

Collectibles, and particularly art, have long been of interest to economists due to their potential value as alternative investments (Burton & Jacobsen 1999; Gérard-Varet 1995). What makes a painting or sculpture "good" is fundamentally subjective, yet there is some agreement about the visual appeal of certain pieces of art and the emotions they arouse in the viewer. A print or reproduction of a piece may reproduce the aesthetics of the original, but it lacks the uniqueness, scarcity, and associated status signal to make it a potential investment.

It is unsurprising, then, that the death of an artist is typically associated with an increase in the market price of their works (e.g., Ekelund et al. 2000; Etro & Stepanova 2021), since the "death effect" can be explained in economic terms as the result of supply of the artist's works becoming fixed (i.e., providing some certainty about the expected future supply of works by the artist).[6] For example, Pénasse et al. (2021) show that the effect of a premature death is particularly large (55%), which is consistent with the artist's death providing relevant information about the future supply of the artist's works; Etro and Stepanova (2021) find a similar pattern. Ursprung and Wiermann (2011) point out that these results are sensitive to the sampling frame and model specification in the sense that unobserved reputation may confound the estimated coefficient on age at death.

Another possibility is that media attention surrounding the death of an artist provides information that reminds or informs collectors of an artist's stature and oeuvre. Matheson and Baade (2004) refer to this as a "nostalgia effect," which would effectively increase demand for the artist's work and, like the reduction in expected future supply, put upward pressure on prices. More generally, consumers' decisions are influenced by their mood and emotions (Lowenstein 2000; Hsee & Rottenstriech 2004), which could be affected by an artist's death.

To disentangle the two channels through which the death effect on painting prices may operate, Matheson and Baade (2004) instead estimate the death effect on baseball cards, which are printed in fixed print runs each year such that supply is known and determined well before the player passes away.[7] Similar evidence of a "nostalgia" or informational effect of an artist's

[6] That an artists' works appreciate following their death is often considered common knowledge; for example, the character George's decision to invest in a painting by an artist he thinks is dying is a running subplot in the famous "Junior Mint" episode of the television sitcom Seinfeld, which originally aired on March 18, 1993.
[7] Baseball cards are 2.5" by 3.5" cardboard with a photo of the player on one side and some statistics and information about the player on the back. Modern baseball cards originated in the 1952 Topps set, though have their roots in the tobacco cards and trade cards of the mid 1800s. See Jamieson (2010) for a history.

death comes in the form of increased book sales (and increased probability of appearing on the bestsellers list) (Ponzo & Scoppa 2022) and increased album sales (Brandes et al. 2016). Moreover, Brandes et al. (2016) find that the death effect on musicians' album sales is largest for their best-known works, which they interpret as evidence that the information attracts new customers rather than reengaging existing fans.

Together, these studies of artists, authors, ballplayers, and musicians provide compelling evidence that (i) there is a celebrity "death effect" on consumer demand for items related to the celebrity and (ii) that this effect is due to *both* a reduction in future supply of such items (supply shock) and an increase in demand for such items (nostalgia effect). The second point here is nuanced: even when there is no *direct* supply shock caused by a celebrity death for memorabilia that is already of a fixed quantity (e.g., baseball cards, event ticket stubs), demand for such products may nonetheless be affected by the sudden shock to substitute goods that do suddenly become of fixed supply (e.g., autographs, in-person photographs with the celebrity).

However, a puzzle arises in the case of trading cards, which enjoy an active and thick market: why do savvy investors not exploit this predictable price bump by purchasing the key cards of legendary players while they are still living? The current study estimates the size and persistence of the death effect on baseball hall-of-famers' rookie cards' market prices by extending Matheson and Baade (2004), the only existing study of death effects on investment grade collectibles of which we are aware, in three important ways.[8]

First, we improve on Matheson and Baade by using more detailed data. Specifically, we use actual market-clearing price data from confirmed online sales and auctions rather than price-

[8] Note that we only seek to credibly estimate the magnitude and persistence of such effects. Separately identifying and distinguishing between mechanisms, such as nostalgia-induced demand shocks and changes in the availability of substitute goods, is beyond the scope of this paper.

guide data. This has the advantage of tracking actual sale prices on a daily basis as opposed to somewhat arbitrary and opaquely sourced price guide projections that are released at regular intervals. Additionally, we focus solely on sales of PSA-graded cards, which ensures we know the quantity available of each card in each grade and the condition of cards being sold, the latter of which is particularly important for older "vintage" cards.[9]

Second, we analyze an objectively defined analytic sample of nine players who had a unique and agreed upon key rookie card and whose death occurred during a timeframe in which a large number of transactions were recorded both before and after the death.

Finally, as described in greater detail in Section 4, we improve on the empirics in Matheson and Baade (2004) by adopting a quasi-experimental event-study research design that exploits actual sales observed both before and after the player's death. This research design adjusts for pre-existing trends in both the card market overall and in the specific card, player characteristics, and card characteristics.

3. DATA

The analytic sample contains all verified sales of PSA-graded Topps rookie cards of the nine players summarized in Table 1, which were issued between 1957 and 1967. Sales data were manually pulled from the PSA website.[10] We identified the nine players as follows. First, we focus on members of the Major League Baseball (MLB) Hall of Fame. This is an objective measure of a player's importance and prominence. Second, we limit attention to the years that

[9] Professional Sports Authenticator (PSA) is a leading grader of sports cards. They grade cards on a 10-point scale and encase them in a protective, serial-numbered holder that verifies the card's authenticity and condition. PSA makes the graded population count of each card publicly available alongside each verified sale price. See https://www.psacard.com/ for additional information.

[10] https://www.psacard.com/auctionprices

Topps had a monopoly on the production of MLB-licensed trading cards (1956-1981). During these years each player has a unique rookie card that is the key card of that player. Third, we restrict attention to players who passed away between 2017 and 2021. This ensures that there are 270 days (three quarters) of reliable sales data both before and after the passing of each player. The availability of reliable sales data aligns with the widespread adoption of the online auction house Ebay by trading card collectors in 2016.

Figure 1 provides a histogram and kernel density estimate of the distribution of sales frequencies relative to the player's death. Aside from a notable spike in transactions in the days immediately following the player's death, the distribution is approximately uniform, averaging about five sales per day.

Table 1 shows that the sample is diverse in several respects. It contains five pitchers and four position players. Position players include "slugging" outfielders Willie McCovey and Frank Robinson, speedster Lou Brock, and infielder Joe Morgan. It contains five Black players and four white players. All four white players are pitchers, and all four position players are Black. Seven MLB teams are represented.

Column 1 of Table 2 summarizes the analytic sample at the transaction level, which is the unit of analysis for the main interrupted time series regression analyses. A total of 2,161 sales of PSA-graded rookie cards of these nine players occurred between September 2016 and December 2021. The average sale price was $1,291, and sales were approximately evenly distributed before and after the player's death. Sales of three players, all pitchers (Sutton, Bunning, and Niekro), are significantly less common than sales of the others. The average grade was 5.8 (out of 10).[11]

[11] Cards with non-numerical grades, including autographed and altered cards, are excluded from the sample.

Forty-two percent of sales took place in 2020, which is consistent with a general spike in interest in trading cards during the initial Covid-19 lockdowns (e.g., Channick 2021).

Columns 2 and 3 of Table 2 summarize the analytic sample separately for pre- and post-death sales, respectively. The average post-death sale price was $1,310 greater than the pre-death average. This may be partly explained by the higher average grade of cards sold post-death. The frequency of sales of individual players' cards, however, was similar pre- and post-death. No 2016 sales are post-death, and no 2021 sales are pre-death, though Don Sutton died in early 2021.

Finally, columns 4 and 5 of Table 2 summarize the analytic sample separately for sales of white and Black players' cards, respectively. Racial disparities (biases) in baseball card prices are documented in some instances (e.g., Fort & Gill 2000) and a similar pattern is observed here in the raw data. This could be conflated with the fact that the four white players are all pitchers and pitchers' cards might sell at a premium (Fort & Gill 2000). Alternatively, it could be due to the white players' cards having higher grades, on average, than those of the Black players. Tom Seaver is the most frequently sold white player at nearly twice the rate of his counterparts; Black players' sales are more evenly distributed.

4. METHODS

We attempt to identify the causal effect of a player's death on the market price of their rookie baseball cards using an interrupted time series (ITS) design (e.g., Morgan & Winship 2015, pp. 244 – 248). We estimate these models using the pooled analytic sample and separately for each player. Intuitively, the ITS design attempts to identify a structural break or discontinuity in the trend in prices by controlling flexibly for a function of time relative to the interruption,

which in this case is the player's death (*DAYS*). Pooled models also control for player fixed effects (ω), which make controlling for individual player attributes or statistics redundant. All models control for specific aspects of the transaction (X).

Specifically, we estimate models of the form:

$$\log(p_{si}) = \beta POST_{si} + f(DAYS_i) + \gamma X_s + \omega_i + \varepsilon_{si}, \quad (1)$$

where *s* and *i* index sales and players, respectively. The natural log of prices is used to normalize the distribution of prices and so that the parameter of interest, β, represents a semi-elasticity. Linear, quadratic, and piece-wise specifications of *f* are considered, as are several different bandwidth restrictions on days relative to the event, in a battery of robustness tests. Standard errors are clustered by transaction date, as this is the level at which treatment varies (Abadie et al. 2023).[12]

Elements of X include controls for the transaction date such as day of week, month, and a full set of quarter indicators. Day-of-week indicators adjust for the likelihood that sales are more common and auctions draw more interest on weekends; there is a parallel in the finance literature known as the "Monday Effect" (French 1980). Similarly, month indicators adjust for seasonal variation in the trading card market due to professional sports' schedules, holiday shopping, and so on; there is a parallel here in the finance literature known as the "January Effect" (Thaler 1987). Quarter indicators account for the general price levels and interest in the trading card market at a given point in time; these are sometimes replaced by a cubic time trend to account for

[12] Clustering at higher levels, such as by transaction quarter, yields smaller (less conservative) standard errors. Clustering by player would be overly conservative given the small number of players and is approximately accounted for by player FE.

the growth, correction, and recovery in the general card market during this time. Finally, X includes the grade and population of the card involved in the sale.[13]

The identifying assumption in an ITS is that the pre-event trend in the outcome would have continued in the same fashion had the event not occurred. This is plausible in the current context because the exact timing of a player's death is unlikely to be associated with another discrete shock to trading card prices. We provide corroborating evidence by showing that the main results are robust to controlling for calendar time (i.e., aggregate shocks to the trading card market) and to the choice of functional form for f.

5. RESULTS

5.1 Main Results

Baseline estimates of equation (1) are presented in Table 3. Because sale prices are logged the estimated coefficient on the Post-Death indicator (β) can be interpreted as a semi-elasticity. Each column of Table 3 estimates equation (1) using a different specification of $f(DAYS)$ and/or a different functional form for the transaction date. Column 1 uses a simple linear control for days relative to the player's death and a full set of quarter-of-transaction FE. The estimated semi-elasticity is 0.20, which suggests that a player's death roughly causes a 20% increase in sale price.[14] Controls for the card's grade and population have the expected signs:

[13] The OLS estimate of β is robust to functional form through which the grade and population variables enter the model. Of particular interest is that both nonlinear and nonparametric functions of grade show increasing returns to grade, though we leave an in-depth analysis of this relationship to future work. Another concern involving grades is that card quality is continuous while grades are discrete, and thus within a given grade some cards may look better than others. It is possible, then, even conditional on grade, the quality of cards for sale improves following a player's death due to an increase in the size of the market. A suggestive test for this hypothesis amounts to estimating equation (1) using card grade as the outcome variable to see if the mix of cards being sold changes. We find no evidence that this is so, suggesting that the results are not driven by a change in the unobservable quality of the cards being sold post death. Results are available upon request.

[14] Using the exact formula of $100(e^{\beta} - 1)$ suggests an increase of 22%.

cards with higher grades (i.e., that are in better condition) sell for significantly more than those with lower grades, and those with higher populations (i.e., those that are less scarce) sell for significantly less than those with smaller populations.

The latter result is not necessarily a causal effect because of selection into the grading process: a card in obviously poor condition is less likely to be graded since grading is costly. This means that population counts for low-grade cards are artificially low. A careful analysis of the relationship between grades, population counts, and prices is warranted but falls outside the scope of this study. We leave this to future work.

The estimated player fixed effects (FE) show that all eight players sold at a discount relative to Bob Gibson, who was arbitrarily chosen as the omitted reference category (due to alphabetical order of first names). Don Sutton, Phil Niekro, and Jim Bunning sold for approximately 90% less than Gibson. This could be due to Gibson being a more dominant pitcher than Sutton and Bunning, as evidenced by his higher wins above replacement (WAR) statistic in Table 1, and his rookie card being 5 years earlier than Niekro's.[15] He is also arguably a more iconic player due to a truly dominant run in the 1960s that included two World Series Championships, two Cy Young Awards for best pitcher, and one Most Valuable Player (MVP) award, alongside his prominent role in the Civil Rights Movement (Gibson & Wheeler 1996).

Columns 2 through 4 of Table 3 augment this simple specification to control more richly for days-to-event. The most flexible of these, the piecewise-quadratic specification in column 4, yields a semi-elasticity of about 0.18, and all yield estimates between 0.18 and 0.20, suggesting that the estimated death effect is robust to how $f(DAYS)$ is specified. This is reassuring, as the ITS is based on the idea that an extrapolation of the pre-event trend provides a good estimate of

[15] The WAR statistic facilitates holistic performance comparisons of players of the same or different positions. "Replacement" refers to relative value over an average player at the same position.

the counterfactual trend. The negative coefficient on Post*Days in column 3 suggests that the death effect deteriorates over time and approaches zero approximately 100 days post-death.

Columns 5 and 6 re-estimate the linear and quadratic piecewise models, respectively, but using a cubic time trend in place of the quarter-of-transaction indicators. This parametric approach to controlling for transaction date is more parsimonious but assumes smooth trends in overall price levels in the market. Controlling for transaction date is important given the growth, correction, and recovery in trading card values between 2017 and 2021. Once again, it is reassuring that the estimated semi-elasticity is robust to this modeling decision, which suggests that the estimated death effect is not due to player deaths occurring at times when overall card values were rising. Moving forward, we focus on the piecewise-linear specification that controls nonparametrically for quarter of transaction because the piecewise-quadratic interaction terms are not statistically significant, nor are the individual elements of the cubic time trend.

Table 4 provides another set of sensitivity analyses: re-estimating the baseline piecewise-linear specification for different samples that are restricted to sales that take place within an increasingly tighter window around the player's death. As in the common practice of experimenting with different bandwidths in regression discontinuity designs (e.g., Lee & Lemieux 2010), this exercise is motivated by the idea that a tighter window around the player's death compares sales that are occurring in similar economic and market conditions. The baseline estimates use sales that occurred within 270 days on either side of the player's death and are therefore spread out over about a year and a half; these estimates are reproduced in column 1. Columns 2 through 6 present estimates from successively tighter windows (i.e., smaller samples), culminating with estimates from a sample of 181 transactions that occurred within a 30-day period that includes the two weeks before and the two weeks after each player's death.

The 180-day restriction provides a nearly identical estimate of about 0.18. The 90- through 15-day samples yield larger estimates ranging from 0.28 to 0.31; all are statistically significant.

The estimates in Table 4 are interesting for two reasons. First, that they are qualitatively similar to one another suggests that the main finding of a death bump in prices is not an artifact of any one particular time period or sample of sales. Second, that estimates from the smaller bandwidths are larger than those in the full sample suggest that the immediate increase in prices is larger, by as much as 20%, and then fades out. Still, prices remain elevated relative to those implied by the pre-death trend for at least 60 days. This fade out can be seen by the negative coefficient on the Post*Days term in the piecewise-linear specifications, which is strongly significant only in the larger samples, and is depicted in Appendix Figure A.1.

We also conduct a randomization inference "placebo" exercise to ensure that we are identifying real death effects and not spurious price increases, given the dramatic price increases generally observed during this time frame (Athey & Imbens 2017; Hess 2017; Young 2018). Specifically, we randomly assign a placebo death date to each player, re-estimate the baseline model given in equation (1), and save the estimated coefficient on POST. We repeat this process 1,000 times and then plot the distribution of estimated βs in Figure 2. These placebo estimates should be zero, as the dates in these regressions are meaningless. This is precisely what we see in Figure 2: the distribution is tightly centered on zero, most estimates are between -0.05 and 0.05, and all estimates are smaller than the baseline estimate of about 0.20. This last fact implies an empirical p value of zero, which is consistent with the baseline estimates in Table 3, and suggests that the significant findings in Table 3 are not due to chance.

Figure 3 provides a final robustness check in the form of fully non-parametric estimates of the death effect for a model that replaces the post-event indicator and parametric function of

days-to-event with an indicator for each sale date observed in the +/- 30-day sample. These estimates are plotted with their 95% confidence intervals and show a clear and persistent level shift post-death of the same magnitude identified in column 5 of Table 4. Overall, across a variety of different samples and model specifications, players' deaths create a robust, statistically significant, and persistent increase in their rookie card valuations.

5.2 Heterogeneity

The estimated player FE reported in Table 3 highlight the level differences in average sale prices of the nine players' Topps rookie cards. This raises the possibility that the death effect itself also varies across players or by player type. To examine the former, we estimate the baseline model separately for each player. These estimates are reported in Figure 4 and Appendix Table A1. Most are in the 0.2 to 0.5 range, and five are statistically significant at the 95% confidence level. Bob Gibson yields a large but imprecisely estimated effect. The estimates for Tom Seaver and Jim Bunning are smaller at about 0.10 and are not statistically significant at traditional confidence levels. Still, the player-specific estimates are uniformly positive and not too different from one another in magnitude, suggesting that while there is some variation in the death effect across players, no one player drives the main results.[16] Lastly, the imprecision of Gibson's and Seaver's could be due to public awareness of the serious illnesses they fought prior to passing.

We also investigate heterogeneity at the group level, which is motivated by some existing literature and the player-level heterogeneity described above. This analysis provides greater

[16] We formalize this using a "leave-one-out" sensitivity analysis, in which the baseline model is estimated nine times, each time excluding sales of one player from the analytic sample. These estimates are reported in Appendix Table A2. The baseline estimate of about 0.20 is quite robust to omitting any one player from the sample.

power with which to identify heterogeneous effects. These results are presented in Table 5. In columns 1 and 2 we estimate the baseline piecewise-linear specification separately for pitchers and position players, respectively. The regression-adjusted semi-elasticity for position players is about 70% larger than that for pitchers, though both estimates are strongly statistically significant. This could be because the news shock associated with a player's death is magnified in the larger market for position players or the general prestige of the position players relative to the pitchers included in this particular analytic sample.

A quirk of the analytic sample is that all four position players are Black and all four white players are pitchers. Accordingly, testing for heterogeneity by race will likely yield similar results as the positional analysis, as the only difference is that Bob Gibson enters the Black sample and leaves the otherwise all-white pitcher sample. As expected, the Black semi-elasticity is noticeably larger than the white semi-elasticity, and once again both are strongly statistically significant. This runs counter to the general finding of racial discrimination in collectibles markets in the form of lower pricing for Black artists, players, and figures (Fort & Gill 2000; Nguyen 2022). This could be an example of the position-player premium dominating the discrimination penalty, though a careful analysis of the presence and severity of racial discrimination in the market for graded rookie cards of MLB Hall-of-Famers falls outside the scope of this study and is left for future work.

Finally, recall that four of the players' rookie cards are shared with other, significantly less prolific players. Anecdotally, these multi-player cards are less attractive to collectors, all else equal. In columns 5 and 6 we estimate the model separately for multi- and single-player rookie cards. The estimated semi-elasticity is about 30% larger for solo rookie cards, as expected. This suggests that the card's design and visual appeal moderates the death bump.

6. Conclusion

This paper estimates interrupted time series (ITS) models that document large and statistically significant "death effects" of legendary baseball players' deaths on market valuations of their rookie cards. These effects are arguably causal and on the order of 20%. Effects are largest in the days immediately following the death, but prices remain elevated for at least 60 days. Generally, our findings reaffirm the presence of a causal "nostalgia" or informational effect of celebrity and artist deaths on consumer demand for associated collectibles (Matheson & Baade 2004), music (Brandes et al. 2016), and books (Ponzo & Scoppa 2022).

However, unlike in the case of records and books, collectibles like vintage baseball cards cannot be reprinted or reproduced digitally. Therefore, the presence of a sizable 20% price increase that persists for several weeks, if not months, presents a puzzle: why are savvy collectors and investors not exploiting this profit opportunity by buying the cards of living hall-of-famers? At first blush, this refutes the basic premise of the EMH and adds another datapoint to the literature on pricing anomalies in asset markets. Baseball card collectors, it seems, do not have fully rational expectations… or do they?

There are a few broad reasons that might explain our findings. First, while players' deaths eventually occur with certainty, the timing of those deaths is uncertain. To exploit the arbitrage opportunity, investors would have to buy and hold the cards of living players, which would entail explicit storage costs and implicit opportunity costs. The latter are most relevant here, as money would be tied up in the cards that cannot be used for other types of consumption or investment.

Second, there can be nontrivial transaction costs associated with selling trading cards. One type of cost here relates to shipping the card to its purchaser. Even when the buyer pays for

shipping, which is not included in the sales price we report, the seller still must take time to carefully package the card and ensure it gets to the shipping company. Search and negotiation frictions pose another type of transaction cost, which can be formidable.

A simple way to estimate these transaction costs is to assume that sellers use consignment services to sell their cards. Full-service consigners do everything from listing the card for sale or auction on an internet auction site, paying the relevant listing and seller fees, and shipping the card when it sells. For cards in the price range of those in our analytic sample, consignment fees would typically be between 10 and 15% of the sale price.[17] These fees negate nearly the entire profit opportunity for cards trading under $1,000: a $100 card that sells for $120 following the player's death leaves the investor with only $102, a 2% net return, after paying the 15% consignment fee. More expensive cards present a more lucrative return: a $1,000 card that sells for $1,200 following a player's death leaves the investor with $1080, an 8% net return, after paying the 10% consignment fee. Not as striking as the 20% nostalgia effect estimated in our main specification, but a nontrivial profit, nonetheless.

Moreover, this 8% net return on a $1,000 investment is likely a lower bound on the profit realized from this sort of investing strategy, as postwar vintage baseball card prices were growing significantly during the period of our sample (2016 – 2021); recall that the Card Ladder vintage index shows growth of about 250%. Of course, the stock market was growing during this time as well: an alternative investment in, say, a NASDAQ index fund would also have tripled in value during this timeframe. The difference between these two investments would have been the 8% "extra" return on the cards of the players who died.

[17] For example, see the payout structure of one popular consigner, dcsports87, here: https://dcsports87.com/how-process-works.

So why is this 8% net profit not arbitraged away? It could be that hoping to profit from players' deaths is a morally repugnant strategy that investors shy away from. Alternatively, the practical challenges of buying and storing enough high-value cards of each of the roughly 60 living hall-of-fame players over an unknown time horizon to make the effort worthwhile are insurmountable. In any case, it does appear that for at least some higher value trading cards, a small profit opportunity exists.

Finally, as high-frequency trading-card sales data are now readily available via Ebay and data aggregators such as PSA and Card Ladder, these results suggest that several related questions regarding the determinants of trading card prices – and their implications for economic and consumer theory – could be revisited using credible quasi-experimental designs. For example, the degree of racial discrimination in collectibles markets; the marginal rate of substitution between the scarcity (population count) and other aspects of a given card such as aesthetics or the player's stature; and the size of the player's hometown market as a proxy for demand are just three seemingly open questions that could be addressed. Another question for future work regards the external validity of these findings and more generally how the determinants of collectible pricing vary with market conditions, as the trading card market was quite bullish during the timeframe studied here (2016 – 2021).

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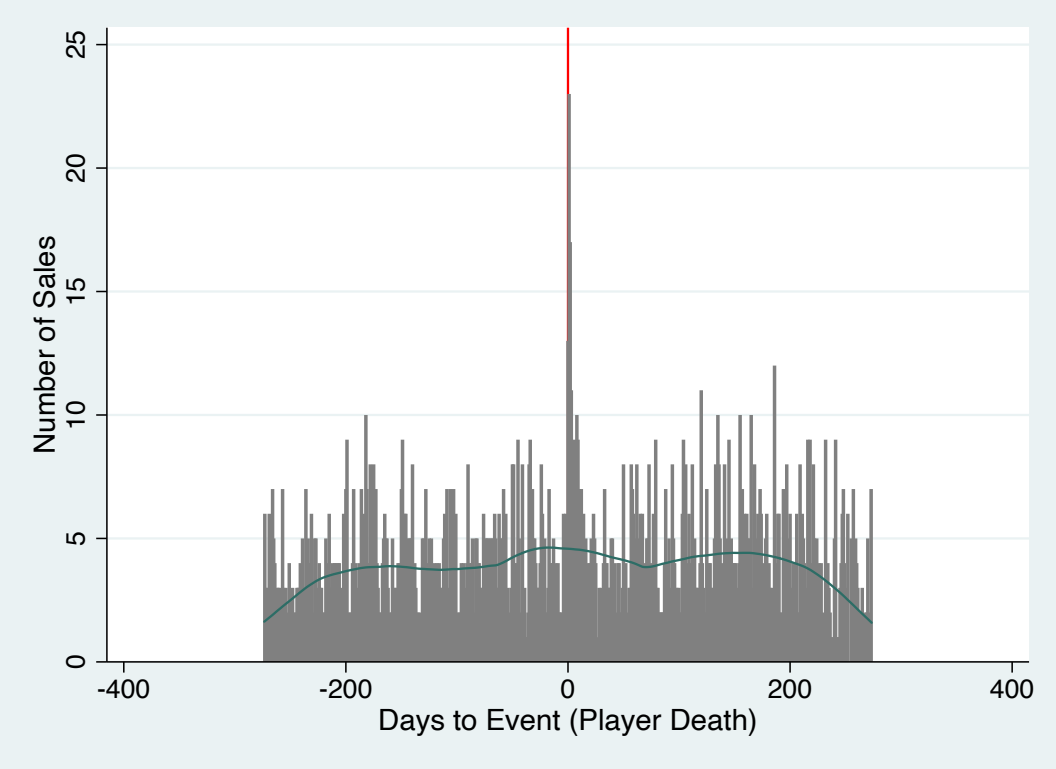
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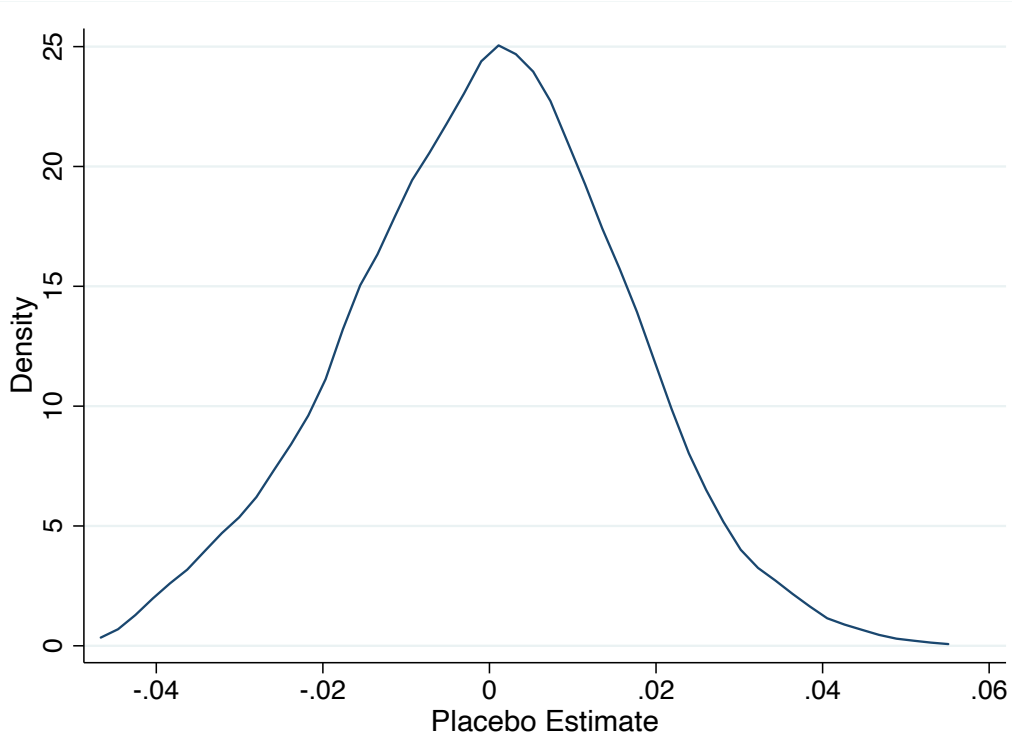
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Figure 1: Distribution of Sales Relative to Date of Player Death



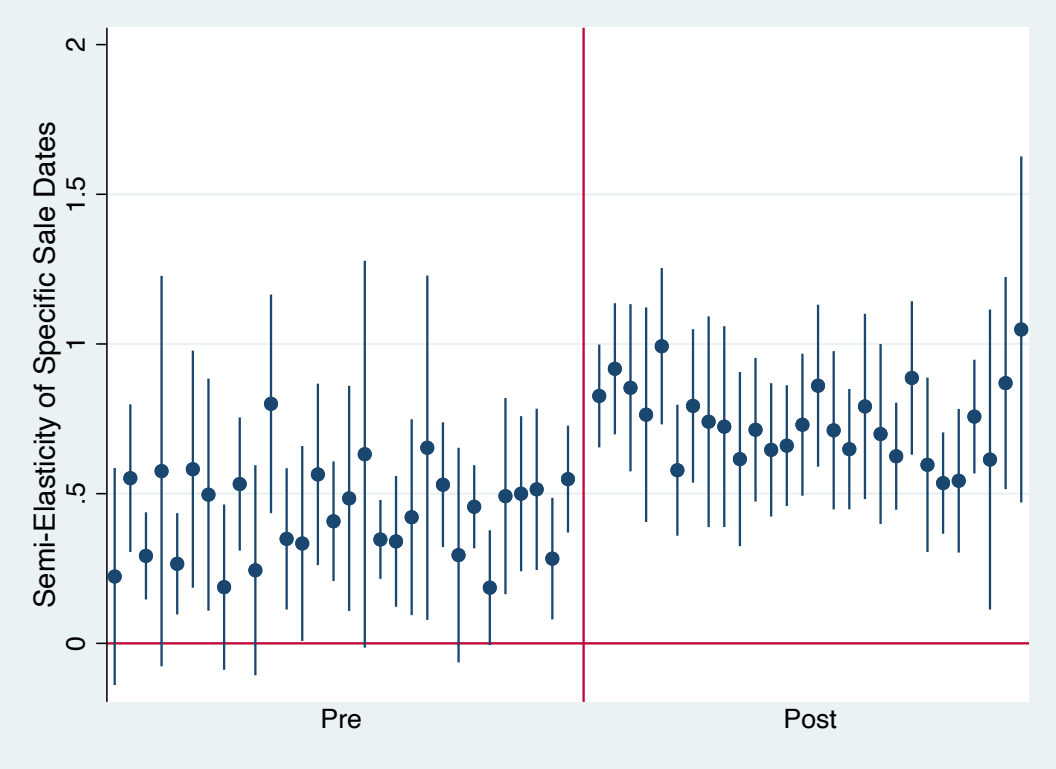
Notes: Day 0 is day of death. Red reference line is at 0.

Figure 2: Placebo Estimates from Randomization Inference



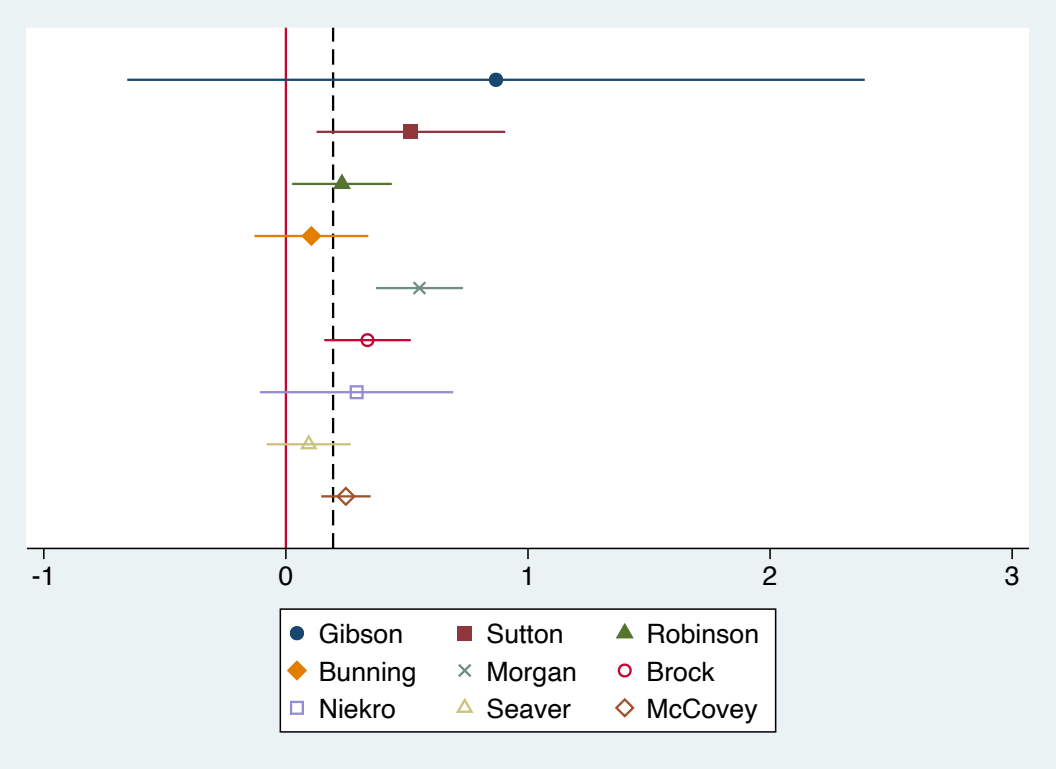
Notes: Kernel density estimate of distribution of estimated placebo effects from randomization inference with 1,000 replications using the preferred baseline model of column 3, Table 3. Placebo estimates are of β in equation (1).

Figure 3: Non-Parametric Regression Estimates



Notes: Estimates from a fully nonparametric specification of *f*(Days) Pre- and Post-Death. N = 295 transactions within 30 days on either side of the player's death. The regression controls for day-of-week fixed effects (FE), month FE, quarter, and player FE, as well as card grade and card population. 95% confidence intervals are clustered by transaction date (N = 144).

Figure 4: Player-Specific Death Effect Semi-Elasticities



Notes: The dependent variable is the logged sale price. Nine separate regressions were estimated for sales of each of the nine players' cards. Coefficient estimates and standard errors are reported in Appendix Table A1. All regressions use the piecewise-linear functional form and control for day-of-week fixed effects (FE), month FE, and quarter FE as well as controls for the card's grade and population. Error bars are clustered by transaction date for the 95% confidence interval. The dashed line indicates the full sample estimate of 0.195.

Table 1: Players in Analytic Sample

Player	Rookie Year (1)	Inducted (2)	Died (3)	Position (4)	Race (5)	HOF Team (6)	WAR (7)	Wins / HR (8)
Bob Gibson	1959	1981	2020	P	Black	STL Cardinals	89.2	251
Don Sutton	1966 M	1998	2021	P	White	LA Dodgers	66.7	324
Frank Robinson	1957	1982	2019	RF	Black	Cincinnati Reds	107.2	586
Jim Bunning	1957	1996	2017	P	White	Detroit Tigers	59.5	224
Joe Morgan	1965 M	1990	2020	2B	Black	Cincinnati Reds	100.4	268
Lou Brock	1962	1985	2020	LF	Black	STL Cardinals	45.3	149
Phil Niekro	1964 M	1997	2020	P	White	Atlanta Braves	95.9	318
Tom Seaver	1967 M	1992	2020	P	White	NY Mets	109.9	311
Willie McCovey	1960	1986	2018	1B	Black	SF Giants	64.5	521

Notes: An M in column 1 indicates that the player's rookie card was a multi-player card; for example, Tom Seaver's 1967 pictures both he and Bill Denehy, a significantly less accomplished player who was also a rookie on the 1967 Mets. Column 6 reports the team the player was inducted into the Hall of Fame (HOF) as a member of. WAR is the advanced statistic "Wins Above Replacement." HR is career home runs. Column 8 reports HR for position players and wins for pitchers.

Table 2: Summary Statistics

	All	Pre	Post	White	Black
	(1)	(2)	(3)	(4)	(5)
Log(Price)	5.78	5.50	6.03	5.99	5.69
	(1.35)	(1.24)	(1.39)	(1.50)	(1.27)
Price	$1291	$597	$1907	$1676	$1123
	(10924)	(1587)	(14908)	(13651)	(9490)
Days to death	4.33	-133.32	126.48	19.15	-2.14
	(151.71)	(77.63)	(79.68)	(151.75)	(151.29)
After death	0.53	0.00	1.00	0.58	0.51
Bob Gibson	0.17	0.17	0.17	0.00	0.24
Don Sutton	0.07	0.06	0.07	0.22	0.00
Frank Robinson	0.13	0.12	0.15	0.00	0.19
Jim Bunning	0.06	0.05	0.08	0.21	0.00
Joe Morgan	0.12	0.14	0.10	0.00	0.17
Lou Brock	0.13	0.14	0.12	0.00	0.19
Phil Niekro	0.06	0.06	0.06	0.19	0.00
Tom Seaver	0.12	0.10	0.13	0.39	0.00
Willie McCovey	0.14	0.16	0.13	0.00	0.20
Grade	5.78	5.71	5.85	6.33	5.54
	(1.74)	(1.77)	(1.72)	(1.72)	(1.70)
PSA Population	616	613	618	407	707
	(290)	(287)	(292)	(174)	(283)
2016	0.01	0.03	0.00	0.04	0.00
2017	0.04	0.02	0.06	0.15	0.00
2018	0.13	0.25	0.03	0.02	0.18
2019	0.16	0.04	0.26	0.01	0.22
2020	0.42	0.66	0.22	0.44	0.42
2021	0.23	0.00	0.43	0.34	0.18
N	2161	1016	1145	657	1504

Notes: This table summarizes the main analytic sample. The unit of analysis is individual recorded sales of PSA-graded cards that took place between 2016 and 2021 and within 275 days of the player's death. Dates refer to the date of the sale and how they relate to the date of the player's death. Grades PSA grades on a 1-10 integer scale. Population is the total number of identical cards that have received the same grade from PSA as of December 2022. Columns 2 and 3 report means separately for transactions that occurred before and after the player's passing, respectively. SD are reported in parentheses for non-binary variables.

Table 3: Regressions of Logged Sale Price using the Pooled Sample

f(*Days*)	Linear	Quadratic	Piecewise Linear	Piecewise Quadratic	Piecewise Linear	Piecewise Quadratic
	(1)	(2)	(3)	(4)	(5)	(6)
Post	0.202	0.201	0.195	0.178	0.171	0.174
	(0.038)***	(0.038)***	(0.038)***	(0.045)***	(0.035)***	(0.047)***
Days	0.001	0.001	0.002	0.003	-0.001	-0.000
	(0.001)	(0.001)	(0.001)**	(0.001)***	(0.000)	(0.001)
$Days^2$		-0.000		0.000		0.000
		(0.000)***		(0.000)**		(0.000)
Post*Days			-0.002	-0.004	-0.001	-0.002
			(0.000)***	(0.001)***	(0.000)***	(0.001)**
$Post*Days^2$				-0.000		0.000
				(0.000)		(0.000)
Grade	0.532	0.533	0.533	0.533	0.533	0.533
	(0.008)***	(0.008)***	(0.008)***	(0.008)***	(0.008)***	(0.008)***
Population	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
	(0.000)***	(0.000)***	(0.000)***	(0.000)***	(0.000)***	(0.000)***
Gibson	(omitted)					
Sutton	-2.796	-2.793	-2.791	-2.791	-3.04	-3.044
	(0.114)***	(0.113)***	(0.113)***	(0.112)***	(0.066)***	(0.068)***
Robinson	-1.251	-1.299	-1.308	-1.282	0.109	0.139
	(0.610)**	(0.608)**	(0.609)**	(0.609)**	(0.237)	(0.245)
Bunning	-3.615	-3.705	-3.735	-3.704	-1.140	-1.079
	(1.196)***	(1.191)***	(1.192)***	(1.196)***	(0.532)**	(0.557)*
Morgan	-2.038	-2.040	-2.039	-2.037	-2.057	-2.057
	(0.033)***	(0.033)***	(0.033)***	(0.033)***	(0.032)***	(0.032)***
Brock	-1.360	-1.364	-1.367	-1.368	-1.311	-1.311
	(0.038)***	(0.038)***	(0.038)***	(0.038)***	(0.029)***	(0.029)***
Niekro	-2.467	-2.462	-2.458	-2.457	-2.655	-2.657
	(0.104)***	(0.104)***	(0.104)***	(0.104)***	(0.069)***	(0.071)***
Seaver	-0.140	-0.143	-0.146	-0.148	-0.081	-0.081
	(0.047)***	(0.047)***	(0.047)***	(0.047)***	(0.040)**	(0.040)**
McCovey	-1.973	-2.027	-2.034	-2.001	-0.403	-0.366
	(0.701)***	(0.698)***	(0.699)***	(0.700)***	(0.279)	(0.289)
Quarter FE	×	×	×	×		
Cubic(date)					×	×

Notes: N = 2,161 transactions that occurred within +/– 270 days of the player's death. All regressions control for day-of-week fixed effects (FE), month FE, and player FE, where Bob Gibson is the omitted reference player. Columns 5 and 6 replace quarter-of-transaction FE with a cubic time trend. Standard errors are clustered by transaction date (N = 911) and reported in parentheses.

Table 4: Robustness to Transaction Date Range about Player Death

Bandwidth	+/– 270	+/– 180	+/– 90	+/– 60	+/– 30	+/– 15
	(1)	(2)	(3)	(4)	(5)	(6)
Post	0.195	0.180	0.277	0.309	0.258	0.313
	(0.038)***	(0.040)***	(0.052)***	(0.060)***	(0.091)***	(0.101)***
Days	0.002	0.002	0.003	0.003	0.004	0.001
	(0.001)**	(0.001)*	(0.002)*	(0.002)	(0.005)	(0.018)
Post*Days	-0.002	-0.002	-0.003	-0.004	-0.006	-0.010
	(0.000)***	(0.001)***	(0.001)**	(0.002)*	(0.005)	(0.019)
Transactions	2,161	1,497	767	532	295	181
Dates	911	638	356	254	144	88

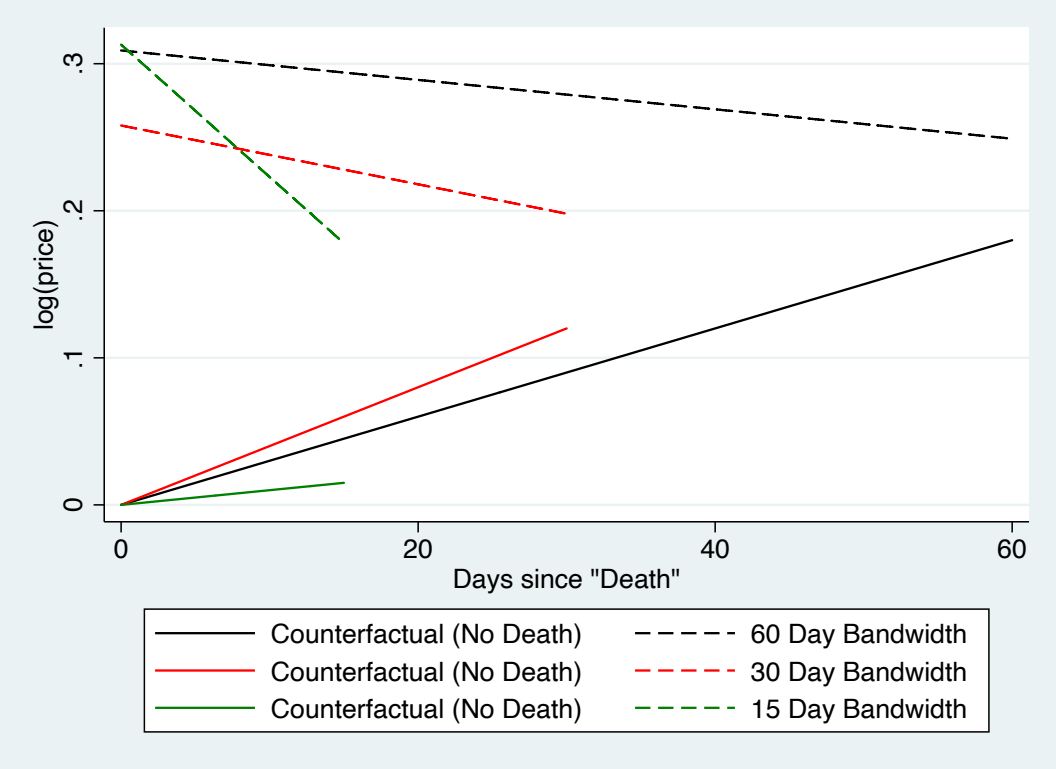
Notes: The dependent variable is the logged sale price. Bandwidth refers to the number of days between the transaction and the player's death. All regressions use the piecewise-linear functional form and control for day-of-week fixed effects (FE), month FE, quarter FE, and player FE as well as controls for the card's grade and population. Standard errors are clustered by transaction date and reported in parentheses.

Table 5: Heterogeneity Analyses

Sample	Pitcher	Position Player	White	Black	Multi-player RC	Solo RC
	(1)	(2)	(3)	(4)	(5)	(6)
Post	0.152	0.261	0.144	0.258	0.180	0.227
	(0.052)***	(0.045)***	(0.048)***	(0.046)***	(0.053)***	(0.049)***
Days	0.001	0.002	0.004	0.001	0.004	0.001
	(0.001)	(0.001)*	(0.001)**	(0.001)	(0.002)**	(0.001)
Post*Days	-0.001	-0.002	-0.001	-0.002	-0.002	-0.001
	(0.001)**	(0.000)***	(0.001)**	(0.000)***	(0.001)***	(0.000)**
Players	5	4	4	5	4	5
Transactions	1,021	1,140	657	1,504	780	1,381
Dates	527	611	415	709	374	766

Notes: The dependent variable is the logged sale price. All regressions use the piecewise-linear functional form and control for day-of-week fixed effects (FE), month FE, quarter FE, and player FE as well as controls for the card's grade and population. Standard errors are clustered by transaction date and reported in parentheses. Differences between the Post coefficients in columns 1 and 2, and between 3 and 4, are significantly different from zero. RC = rookie card.

Appendix Figure A.1: Piecewise Linear Extrapolations of Post-Death Log Prices



Notes: Extrapolations (dashed lines) are based on the estimates in columns 4, 5, and 6 of Table 4. These are estimates of the baseline model presented in equation (4) using bandwidths of 60, 30, and 15 days pre- and post-death, respectively. The "counterfactual" (solid) lines are simple the slopes of the Days coefficients from these models. The vertical axis intercepts of the dashed linear are estimated Death effects, or Post coefficients, from these models.

Appendix Table A1: Player-Specific Regression Estimates

	Gibson	Sutton	Robinson	Bunning	Morgan	Brock	Niekro	Seaver	McCovey
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post	0.868	0.516	0.232	0.105	0.552	0.337	0.292	0.094	0.248
	(0.773)	(0.197)***	(0.104)**	(0.119)	(0.091)***	(0.091)***	(0.201)	(0.088)	(0.052)***
Days	-0.000	0.003	0.008	-0.003	-0.001	-0.001	0.004	0.007	0.001
	(0.002)	(0.004)	(0.003)***	(0.003)	(0.002)	(0.002)	(0.003)	(0.003)**	(0.002)
Post*Days	-0.002	-0.004	-0.003	0.005	-0.001	0.001	0.000	-0.005	-0.000
	(0.002)	(0.003)	(0.002)*	(0.002)***	(0.002)	(0.001)	(0.002)	(0.004)	(0.002)
N	364	142	289	137	260	285	122	256	306
Dates	237	115	185	110	179	199	101	174	214

Notes: The dependent variable is the logged sale price. Nine player-specific regressions are estimated. All regressions use the piecewise-linear functional form and control for day-of-week fixed effects (FE), month FE, and quarter FE as well as controls for the card's grade and population. Standard errors are clustered by transaction date and reported in parentheses.

Appendix Table A2: Leave-One-Out Regression Estimates

	Gibson	Sutton	Robinson	Bunning	Morgan	Brock	Niekro	Seaver	McCovey
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post	0.201	0.189	0.201	0.202	0.185	0.165	0.212	0.216	0.201
	(0.038)***	(0.038)***	(0.044)***	(0.039)***	(0.040)***	(0.040)***	(0.041)***	(0.040)***	(0.042)***
Days	0.003	0.002	0.001	0.002	0.002	0.002	0.002	0.001	0.002
	(0.001)***	(0.001)**	(0.001)	(0.001)**	(0.001)**	(0.001)**	(0.001)*	(0.001)	(0.001)*
Post*Days	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002
	(0.000)***	(0.000)***	(0.000)***	(0.000)***	(0.000)***	(0.000)***	(0.000)***	(0.000)***	(0.000)***
N	1,797	2,019	1,872	2,024	1,901	1,876	2,039	1,905	1,855
Dates	851	874	810	804	890	873	893	879	784

Notes: The dependent variable is the logged sale price. Each column reports the regression that omits sales of the player named in the column header. All regressions use the piecewise-linear functional form and control for day-of-week fixed effects (FE), month FE, and quarter FE as well as controls for the card's grade and population. Standard errors are clustered by transaction date and reported in parentheses.